Answers and talk notes

Choose Appropriate Measures

What your student is learning

Tonight we learn which measures go TOGETHER: mean with MAD for symmetric data, median with IQR when there is an outlier.

Answers

Ask your student to explain before you look.

1. **A.** Mean and MAD. For symmetric data with no outlier, use the mean for center and the MAD for spread: Mean and MAD.
2. **B.** The IQR. The median goes with the IQR, because both depend on the order of the values, not on an outlier. So the answer is: The IQR.
3. **C.** Mean and MAD. No value is far from the others, so report the mean and the MAD. So the answer is: Mean and MAD.
4. **D.** 3, 4, 4, 5, 60. The 60 is an outlier, so use the median and IQR for 3, 4, 4, 5, 60.
5. **A.** Median and IQR. The 90 is an outlier, so report the median and the IQR. So the answer is: Median and IQR.
6. **C.** An outlier does not pull them very far, so they still show the typical group. An outlier does not pull the median or IQR very far, so they still show the typical group. So the answer is: An outlier does not pull them very far, so they still show the typical group.

Talk about it

How did the outlier change the mean but not the median?

Home mission

Write five small numbers on sticky notes, like the ages of cousins. Add one huge note, like 100. Put the notes in order. Ask which is fairer to report: the middle note or the mean.

When they are stuck, ask

- Before we compute — is there a value far from all the others?
- Would the mean here describe a typical value, or get dragged?
- Which measure of spread goes with the center you chose?

Try not to say

- Use the mean, it always uses all the data so it is better.
- Center and spread are separate choices, mix and match.
- Just throw the outlier away.